

ITC-101

Examination – June - 2023

Elect. II Sem : CE, AE, EE, EI, EC, CSE, IT, AIADS, CSE(BC), IoT Python Programming

Max. Marks : 60

Min. Marks : 19

Time : 2 Hrs

Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a & b – Objective type/Fill in the blanks/One sentence answer.

Part c – Max 100 words and Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

Q.1 (a) (i) Computer uses which number system to perform calculations and store data: 02
1. Hexadecimal 2. Decimal 3. Octal 4. Binary

(ii) What type of programming is supported by Python:

1. Object-Oriented-Programming
2. Structured Programming
3. Functional Programming
4. All of the above

(b) The correct extension of python file is _____ 02

(c) List the standard data types in python with one example in each. 03

OR

What is a variable in python? What are the rules for naming a variable? Explain by giving examples. 03

(d) Write a Python program to check whether a number is prime or not prime. 05

OR

What are the various arithmetic and logical operators that python support? Write syntax for each operator. 05

Q.2 (a) (i) Which of the following operators has its associativity from right to left? 02
1. + 2. // 3. % 4. **

(ii) What will be the output of the following code:

n=7

c=0

while(n):

if (n>5):

c = c+n-1

n = n-1

else:

break

print(n)

print(c)

1. 5 11

2. 5 9

3. 7 11

4. 5 2

ITC-101

- (b) What will be the output of the following Python code?
for i in range(0,2,1):

```
print("Hello")
```

- (c) Explain the various control statements in Python with syntax.

OR

Write a program to check whether a person is eligible to vote or not. Take age as input from user.

- (d) Write the output of the following code:

```
x=5
while (x<15):
    print(x**2)
    x+=3
```

OR

Write a program to print the table of a number accepted from user.

- Q.3 (a) (i) What will be the output of:

```
str1 = "PYthon class"
print(str1[1:4],str1[0:-1])
```

1. Yth PYthon clas
2. PYt Python class
3. PYth Python clas

- (ii) What will be the output of:

```
str1= "Python Lab"
print(len(str1))
```

1. 9 2. 10 3. 11 4. 12

- (b) Write the output of the following:

```
str1="PYTHON"
Print(str1.lower())
```

- (c) How to split strings and what function is used to perform that operation?

OR

Write syntax to take a input from a user and typecast it to an integer.

- (d) What is a directory? Which module in python allows to use various methods with directory. Write the syntax for the following: (i) to get current directory in Python (ii) change directory in python (iii) making a new directory in python

OR

Explain the various functions for reading data from a text file.

a) What is the output : 02
colors= ["red", "blue", "green", "pink"]
Print(colors[2])

b) What is the output : 02
colors= ["red", "blue", "green", "pink"]
Print("yellow" in colors)

c) What is the difference between tuples and list? 03

OR

Explain how to create a dictionary in python. 03

(d) Explain indexing and negative indexing in tuple with example. 05

OR

Find the output of the following: 05

```
L1 = [100,900,300,400,500]
```

```
START = 1
```

```
SUM = 0
```

```
for C in range(START,4):
```

```
    SUM = SUM + L1[C]
```

```
    print(C, ":", SUM)
```

```
    SUM = SUM + L1[0]*10
```

```
    print(SUM)
```

Q.5 (a) What is the output of the following code: 02
class Student:

```
    def __init__(self,name,id):
```

```
        self.name=name
```

```
        self.id=id
```

```
        print(self.id)
```

```
std=Student("Simon",1)
```

```
std.id=2
```

```
print(std.id)
```

(b) In python, what is method inside class? 02
1. Attribute 2. Object 3. Argument 4. function

- (c) Explain operator overloading with an example in python.

OR

What is the output of the following code:

```
class A():  
    def disp(self):  
        print("A disp()")  
class B(A):  
    pass  
obj = B()  
obj.disp()
```

- (d) Explain exception handling with example.

OR

Explain Method Resolution order in Python with an example.
